Advanced Fire Fighting (AFF) Exit Exam Questions & Answers PDF

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180 Practice Questions and Answers for STCW Exams

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Practice Set 1

Q1. What is the primary purpose of boundary cooling during a fire on a ship?

A) To minimize smoke production
B) To prevent fire spread by cooling adjacent surfaces
C) To increase the temperature inside the compartment
D) To accelerate the combustion process

Correct Answer: B (To prevent fire spread by cooling adjacent surfaces)

Q2. In advanced firefighting, what is the purpose of using a Class A foam?

A) To cool down hot surfaces
B) To suppress flammable gas release
C) To control oil and fuel fires
D) To neutralize acidic fumes

Correct Answer: C (To control oil and fuel fires)

Q3. During a firefighting operation, what does the term "smoke curtain" refer to?

A) A physical barrier to prevent the entry of firefighters
B) A cloud of water droplets to control temperature
C) A curtain made of fire-resistant material to contain smoke
D) A warning signal for evacuation

Correct Answer: C (A curtain made of fire-resistant material to contain smoke)

Q4. Which firefighting equipment is specifically designed for use in confined spaces onboard a ship?

A) Fire hoses
B) Breathing apparatus
C) Foam monitors
D) Fire extinguishers

Correct Answer: B (Breathing apparatus)

Q5. What is the primary purpose of the International Shore Connection (ISC) on a ship?

A) To supply freshwater to the crew
B) To connect with the shore power supply
C) To facilitate shore-based firefighting assistance
D) To enhance communication with nearby vessels

Correct Answer: C (To facilitate shore-based firefighting assistance)

Q6. In advanced firefighting, what is the purpose of using a water fog nozzle?

A) To create a dense fog for cooling purposes
B) To increase the range of water projection
C) To generate a fine mist for fire suppression
D) To create a visual signal for distress

Correct Answer: C (To generate a fine mist for fire suppression)

Q7. What is the role of a fire blanket in firefighting?

A) To smother small fires by cutting off the oxygen supply
B) To cool down hot surfaces in a fire-affected area
C) To create a barrier against radiant heat
D) To provide an escape route for trapped personnel

Correct Answer: A (To smother small fires by cutting off the oxygen supply)

Q8. In the context of firefighting, what does the acronym PASS stand for?

A) Pressure Application Safety System
B) Portable Aqueous Suppression System
C) Pull, Aim, Squeeze, Sweep
D) Primary Air Supply Source

Correct Answer: C (Pull, Aim, Squeeze, Sweep)

Q9. What is the purpose of a fire screen in advanced firefighting?

A) To protect firefighters from radiant heat
B) To separate different compartments for ventilation
C) To extinguish fires through a fine mist
D) To detect flammable gases

Correct Answer: A (To protect firefighters from radiant heat)

Q10. When using a fire hose, what is the recommended nozzle pattern for initial cooling of a fire in a confined space?

A) Straight stream
B) Fog pattern
C) Solid stream
D) Cone spray

Correct Answer: B (Fog pattern)

Q11. In advanced firefighting, what is the primary function of a fire axe?

A) Ventilating smoke from the compartment
B) Cutting through bulkheads and doors
C) Spraying water in a concentrated stream
D) Applying foam to the fire

Correct Answer: B (Cutting through bulkheads and doors)

Q12. What does the term "fire tetrahedron" refer to in the context of firefighting?

A) A four-sided firefighting tool
B) The four elements required for combustion
C) A specialized firefighting drill
D) A geometric shape used in fire suppression

Correct Answer: B (The four elements required for combustion)

Q13. In the event of an electrical fire onboard, which firefighting equipment is specifically designed for use?

A) CO2 (Carbon Dioxide) extinguisher
B) Foam extinguisher
C) Dry powder extinguisher
D) Water mist extinguisher

Correct Answer: A (CO2 (Carbon Dioxide) extinguisher)

Q14. What is the purpose of a fireman's outfit (protective clothing) in advanced firefighting?

A) To identify firefighting personnel
B) To enhance physical agility
C) To provide protection from heat and flame
D) To store firefighting tools

Correct Answer: C (To provide protection from heat and flame)

Q15. In the context of fire safety, what does the acronym "RECAAR" stand for?

A) Rapid Escape and Control Against Radiant heat
B) Rescue, Evacuation, and Containment of Aerial Risks
C) Response, Escape, and Control Against Adverse Reactions
D) Rescue, Evacuation, Containment, Assessment, and Recovery

Correct Answer: D (Rescue, Evacuation, Containment, Assessment, and Recovery)

Q16. What is the purpose of a flame arrester in firefighting?

A) To control the spread of flammable liquids
B) To arrest the progression of flames in a compartment
C) To provide visual cues during firefighting operations
D) To enhance communication among firefighters

Correct Answer: A (To control the spread of flammable liquids)

Q17. During a shipboard fire emergency, what is the primary purpose of the muster list?

A) To account for all personnel and passengers
B) To provide instructions for firefighting techniques
C) To allocate firefighting responsibilities
D) To identify the location of fire hydrants

Correct Answer: A (To account for all personnel and passengers)

Q18. What is the recommended method for handling an injured person during a shipboard firefighting operation?

A) Immediate evacuation to the shore
B) Placing the person in a confined space
C) Moving the person to a designated safe area
D) Administering first aid at the fire location

Correct Answer: C (Moving the person to a designated safe area)

Q19. Which firefighting agent is commonly used to suppress fires involving flammable metals?

A) Water
B) Foam
C) Dry powder
D) CO2 (Carbon Dioxide)

Correct Answer: C (Dry powder)

Q20. What is the purpose of a fire control plan on a ship?

A) To outline emergency escape routes for passengers
B) To specify the locations of fire alarm pull stations
C) To provide guidance for controlling and suppressing fires
D) To identify the ship's navigation routes

Correct Answer: C (To provide guidance for controlling and suppressing fires)

Q21. What is the purpose of a deluge system in shipboard firefighting?

A) To deliver a high-pressure water mist
B) To flood a compartment with water rapidly
C) To control electrical fires with a dry powder
D) To generate foam for flammable liquid fires

Correct Answer: B (To flood a compartment with water rapidly)

Q22. During a shipboard firefighting operation, what is the primary function of a portable gas detector?

A) To locate fire hydrants
B) To identify flammable gases
C) To measure ambient temperature
D) To monitor radio communication

Correct Answer: B (To identify flammable gases)

Q23. In the context of advanced firefighting, what is the purpose of a water curtain?

A) To create a barrier against radiant heat
B) To supply water to firefighting hoses
C) To generate foam for liquid fires
D) To create a visual signal for distress

Correct Answer: A (To create a barrier against radiant heat)

Q24. Which firefighting equipment is commonly used to cut ventilation openings in ship compartments?

A) Fire axe
B) Water fog nozzle
C) Thermal imaging camera
D) Hydraulic rescue tool

Correct Answer: D (Hydraulic rescue tool)

Q25. During a shipboard firefighting operation, what is the primary purpose of a positive pressure breathing apparatus (PPBA)?

A) To supply fresh air to the compartment
B) To prevent smoke from entering the mask
C) To maintain positive pressure inside the mask
D) To enhance communication among firefighters

Correct Answer: C (To maintain positive pressure inside the mask)

Q30. In advanced firefighting, what is the primary purpose of a fire-resistant bulkhead?

A) To provide a visual barrier against smoke
B) To contain the spread of fire within compartments
C) To enhance communication among firefighting teams
D) To store firefighting equipment safely

Correct Answer: B (To contain the spread of fire within compartments)

Practice Set 2

Q1. A safety is provided on the CO2 discharge piping to prevent:

A) Flooding of a space where personnel are present
B) Over pressurization of the CO2 discharge piping
C) Rupture of cylinder due to temperature increase
D) Over pressurization of the space being flooded

Correct Answer: B (Over pressurization of the CO2 discharge piping)

Q2. Apart from red what other color is frequently used for a CO2 extinguisher:

A) Blue
B) Yellow
C) Black
D) Red

Correct Answer: C (Black)

Q3. Flue gases are used directly for inerting?

A) True
B) False

Correct Answer: B (False)

Q4. In fighting a fire in a fuel tank, the first action you should attempt is to:

A) Station someone at the fixed CO2 release controls
B) Top off the tank to force out all vapors
C) Begin transferring the fuel to other tanks
D) Secure all sources of fresh air to the tank

Correct Answer: D (Secure all sources of fresh air to the tank)

Q5. The FIRST course of action in fighting a fire in a cargo or fuel oil tank is to:

A) Spray the tank boundaries with a fire hose to promote cooling
B) Direct a fire hose into the tank and energize the fire main
C) Discharge an initial charge of CO2 with a hand portable extinguisher
D) Secure all openings to the tank

Correct Answer: D (Secure all openings to the tank)

Q6. Three side of fire triangle are fuel, heat and ....?

A) Petrol
B) Gases
C) Oxygen
D) Combustible material

Correct Answer: C (Oxygen)

Q7. Why should not carbon dioxide extinguisher be used in accommodation space:

A) Frostbite risk
B) It can displace the oxygen, causing Asphyxiation
C) Health hazard

Correct Answer: B (It can displace the oxygen, causing Asphyxiation)

Q8. In a fixed CO2 fire extinguishing system where pressure from pilot cylinders is used to release the CO2 from the main bank of cylinders, the number of required pilot cylinders shall be at least:

A) 6
B) 4
C) 3
D) 2

Correct Answer: D (2)

Q9. Fixed CO2 systems would not be used on crew's quarters or:

A) the engine room
B) cargo holds
C) spaces open to the atmosphere
D) the paint locker

Correct Answer: C (spaces open to the atmosphere)

Q10. Your vessel is equipped with a fixed CO2 system and a fire main system. In the event of an electrical fire in the engine room what is the correct procedure for fighting the fire?

A) Evacuate the engine room and use the fire main system.
B) Evacuate the engine room and use the CO2 system.
C) Use the fire main system and evacuate the engine room.
D) Use the CO2 system and evacuate the engine room.

Correct Answer: B (Evacuate the engine room and use the CO_2 system.)

Q11. When pilot cylinder pressure is used as a means to release the CO_2 from a fixed fire extinguishing system consisting of four storage cylinders, the number of pilot cylinders shall be at least:

A) 1
B) 4
C) 3
D) 2

Correct Answer: D (2)

Q12. After extinguishing a paint locker fire using the fixed CO_2 system, the next action is to have the space:

A) doused with water to prevent reflash
B) checked for oxygen content
C) left closed with vents off until all boundaries are cool
D) opened and burned material removed

Correct Answer: C (left closed with vents off until all boundaries are cool)

Q13. In a fixed CO_2 extinguishing system where provision is made for the release of CO_2 by operation of a remote control, provision shall also be made for releasing the CO_2:

A) at the cylinder location
B) from the cargo control station
C) from the bridge
D) from inside the engine room

Correct Answer: A (at the cylinder location)

Q14. Which of the following statements is FALSE, concerning the regulations pertaining to the cylinder room of a fixed CO_2 fire extinguishing system?

A) The temperature of the room should never exceed 130 degreesF.
B) The door must be kept unlocked.
C) The compartment must be properly ventilated.
D) The compartment shall be clearly marked and identifiable.

Correct Answer: B (The door must be kept unlocked.)

Q15. You are releasing carbon dioxide gas (CO_2) into an engine compartment to extinguish a fire. The CO_2 will be most effective if the:

A) air flow to the compartment is increased with blowers
B) compartment is closed and airtight
C) compartment is left open to the air
D) compartment is closed and ventilators are opened

Correct Answer: B (compartment is closed and airtight)

Q16. CO2 cylinders, which protect the small space in which they are stored must:

A) NOT contain more than 200 pounds of CO2
B) have an audible alarm
C) be automatically operated by a heat actuator
D) All of the above

Correct Answer: A (NOT contain more than 200 pounds of CO2)

Q17. In areas where CO2 piping is installed, such piping may not be used for any other purpose EXCEPT:

A) to run the emergency wiring to the space
B) to ventilate the space
C) in connection with the water sprinkler system
D) in connection with the fire-detecting system

Correct Answer: D (in connection with the fire-detecting system)

Q18. Control valves of a CO2 system may be located within the protected space when:

A) an automatic heat-sensing trip is installed
B) the CO2 cylinders are also in the space
C) there is also a control valve outside
D) it is impractical to locate them outside

Correct Answer: C (there is also a control valve outside)

Q19. While you are working in a space, the fixed CO2 system is accidentally activated. You should:

A) make sure all doors and vents are secured
B) retreat to fresh air and ventilate the compartment before returning
C) continue with your work as there is nothing you can do to stop the flow of CO2
D) secure the applicators to preserve the charge in the cylinders

Correct Answer: B (retreat to fresh air and ventilate the compartment before returning)

Q20. Large volumes of carbon dioxide are safe and effective for fighting fires in enclosed spaces, such as in a pumproom, provided that the:

A) ventilation system is kept operating
B) ventilation system is secured and all persons leave the space
C) persons in the space wear damp cloths over their mouths and nostrils
D) persons in the space wear gas masks

Correct Answer: B (ventilation system is secured and all persons leave the space)

Q21. Automatic mechanical ventilation shutdown is required for CO_2 systems protecting the:

A) galley
B) living quarters
C) cargo compartments
D) machinery spaces

Correct Answer: D (machinery spaces)

Q22. Carbon dioxide as a fire fighting agent has which advantage over other agents?

A) It causes minimal damage.
B) It is safer for personnel.
C) It is cheaper.
D) It is most effective on a per unit basis.

Correct Answer: A (It causes minimal damage.)

Q23. CO_2 cylinders forming part of a fixed fire extinguishing system must be pressure tested at least every:

A) 12 years
B) 6 years
C) 2 years
D) year

Correct Answer: A (12 years)

Q24. Fixed carbon dioxide extinguishing systems, for machinery spaces that are normally manned, are actuated by one control to open the stop valve in the line leading to the space, and:

A) three separate controls to release the CO_2
B) two separate controls to release the CO_2
C) a separate control to release the CO_2
D) the same control releasing the CO_2

Correct Answer: B (two separate controls to release the CO_2)

Q25. A fixed carbon dioxide extinguishing system for a machinery space, designed WITHOUT a stop valve in the line leading to the protected space, is actuated by:

A) one control
B) two controls
C) three controls
D) none of the above

Correct Answer: A (one control)

Q26. On cargo vessels, the discharge of the required quantity of carbon dioxide into any 'tight' space shall be completed within:

A) 1 minute
B) 2 minutes
C) 4 minutes
D) 6 minutes

Correct Answer: B (2 minutes)

Q27. Aboard a cargo vessel, the supply of carbon dioxide used in a fixed extinguishing system MUST at least be sufficient for what space:

A) the space requiring the largest amount
B) the engine room and largest cargo space
C) all cargo spaces
D) all the spaces of a vessel

Correct Answer: A (the space requiring the largest amount)

Q28. How often must CO2 systems be inspected to confirm cylinders are within 10% of the stamped full weight of the charge:

A) quarterly
B) semiannually
C) annually
D) biannually

Correct Answer: C (annually)

Q29. In weighing CO2 cylinders, they must be recharged if weight loss exceeds:

A) 10% of weight of charge
B) 20% of weight of charge
C) 15% of weight of full bottle
D) 10% of weight of full bottle

Correct Answer: A (10% of weight of charge)

Q30. Before using a fixed CO2 system to fight an engine room fire, you must:

A) secure the engine room ventilation
B) secure the machinery in the engine room
C) evacuate all engine room personnel
D) All of the above

Correct Answer: D (All of the above)

Q31. The CO_2 flooding system is actuated by a sequence of steps which are:

A) open stop valve, open control valve, trip alarm
B) open bypass valve, break glass, pull handle
C) sound evacuation alarm, pull handle
D) break glass, pull valve, break glass, pull cylinder control

Correct Answer: D (break glass, pull valve, break glass, pull cylinder control)

Q32. Which advantage does dry chemical have over carbon dioxide (CO_2) in firefighting?

A) Compatible with all foam agents
B) More protective against re-flash
C) Cleaner
D) All of the above

Correct Answer: B (More protective against re-flash)

Q33. What is NOT a characteristic of carbon dioxide fire-extinguishing agents?

A) Effective even if ventilation is not shut down
B) Will not deteriorate in storage
C) Non-corrosive
D) Effective on electrical equipment

Correct Answer: A (Effective even if ventilation is not shut down)

Q34. In a fixed carbon dioxide extinguishing system for a machinery space, designed WITH a stop valve in the line leading to the protected space, the flow of CO_2 is established by actuating:

A) three controls
B) two controls
C) one control
D) none of the above

Correct Answer: B (two controls)

Practice Set 3

Q1. An important step in fighting any electrical fire is to:

A) Apply water fog immediately
B) De-energize the circuit
C) Increase ventilation
D) Use chemical foam

Correct Answer: B (De-energize the circuit)

Q2. Which fire detection system is actuated by sensing a heat rise in a compartment?

A) Flame detection system
B) Manual pull stations
C) Automatic fire detection system
D) Smoke obstruction system

Correct Answer: C (Automatic fire detection system)

Q3. Water fog is an effective firefighting agent because it:

A) Has a great cooling ability
B) Breaks the molecular chain reaction chemically
C) Displaces oxygen completely
D) Smothers the fire using dense foam

Correct Answer: A (Has a great cooling ability)

Q4. A fire of escaping liquefied flammable gas is best extinguished by:

A) Flooding with low-velocity fog
B) Stopping the flow of gas
C) Applying large amounts of dry chemical powder
D) Cooling the surrounding storage cylinders

Correct Answer: B (Stopping the flow of gas)

Q5. When combating a Class C fire, which of the following dangers may be present?

A) Spontaneous combustion of scrap metal
B) High expansion foam flashover
C) Toxic fumes from burning insulation or electric shock
D) Exploding fuel-air vapor mixtures

Correct Answer: C (Toxic fumes from burning insulation or electric shock)

Q6. The lowest temperature at which the vapors of a flammable liquid will ignite and cause self-sustained combustion in the presence of a spark or flame is the:

A) Flash Point
B) Ignition Point
C) Fire Point
D) Critical Point

Correct Answer: C (Fire Point)

Q7. The fire protection provided for the propulsion motor and generator of a diesel-electric drive vessel is usually a:

A) Fixed foam system
B) Fixed CO_2 system
C) Water sprinkler system
D) Dry powder station

Correct Answer: B (Fixed CO_2 system)

Q8. Metal fires should be fought using which extinguisher type?

A) Carbon dioxide
B) Aqueous film-forming foam (AFFF)
C) Dry powder
D) Halon gas

Correct Answer: C (Dry powder)

Q9. Which type of portable fire extinguisher is best suited for putting out a Class D fire?

A) Carbon dioxide
B) Water spray
C) Dry powder
D) Dry chemical

Correct Answer: C (Dry powder)

Q10. Class 1 watertight doors are:

A) Hinged type
B) Vertical sliding type
C) Horizontal sliding type
D) Rolling type

Correct Answer: A (Hinged type)

Q11. A fire in a fuel oil settling tank is a:

A) Class A fire
B) Class B fire
C) Class C fire
D) Class D fire

Correct Answer: B (Class B fire)

Q12. The symbols for Fire Control plans are approved by which organization?

A) International Labor Organization (ILO)
B) International Maritime Organization (IMO)
C) National Fire Protection Association (NFPA)
D) International Standards Organization (ISO)

Correct Answer: B (International Maritime Organization (IMO))

Q13. Which dangerous condition should be kept in mind after using a 6.8 kg CO_2 fire extinguisher on a small oil fire on the engine room floor plates?

A) Suffocation inside the main casing
B) Rupture of downstream valves
C) Rekindling of the fire
D) Freezing of mechanical controls

Correct Answer: C (Rekindling of the fire)

Q14. What emergency equipment is NOT found in the crew mess in the view of a Fire Control plan?

A) Portable fire extinguisher
B) Heat sensor
C) Fire axe
D) Emergency escape breathing device (EEBD)

Correct Answer: B (Heat sensor)

Q15. Dry powder fire extinguishers which contain a mixture of graphite and sodium chloride as firefighting agents are generally used to fight which type of fire?

A) Class A wood fire
B) Class B fuel oil fire
C) Class C electrical fire
D) Class D metal fire

Correct Answer: D (Class D metal fire)

Q16. On international voyages, tank ships of 500 gross tons or more are required to have facilities to enable connection on each side of the ship for this fire control equipment:

A) International Shore connection
B) High expansion foam generator
C) Fixed sprinkler bypass valve
D) Emergency bilge pump suction manifold

Correct Answer: A (International Shore connection)

Q17. Mechanical foam used for firefighting is produced by mechanically mixing and agitating:

A) Foam chemical with nitrogen gas
B) Liquid CO2 with water solution
C) Foam chemical with air and water
D) Dry chemical powder with water stream

Correct Answer: C (Foam chemical with air and water)

Q18. What must be mounted at a small passenger vessel's operating station for use by the master and crew?

A) Vessel maintenance record
B) Emergency instructions
C) SOLAS training booklet
D) International code of signals chart

Correct Answer: B (Emergency instructions)

Q19. The fire main system should be flushed with fresh water whenever possible to:

A) Minimize downstream supply pressure
B) Help destroy marine growth
C) Lubricate fire pump impellers
D) Reduce systemic scale formation

Correct Answer: B (Help destroy marine growth)

Q20. The muster list shows each rig hand's muster station, his duties during abandonment, basic instructions, and:

A) All emergency signals
B) Navigation course parameters
C) Next port of call logistics
D) Engine room operating instructions

Correct Answer: A (All emergency signals)

Q21. If a fire ignites in the engine room as a result of a high-pressure fuel oil leak, you should first:

A) Evacuate all personnel immediately
B) Release the fixed CO2 system
C) Shut off the fuel oil supply
D) Direct a stream of water at the flames

Correct Answer: C (Shut off the fuel oil supply)

Q22. If there has been a fire in a closed, unventilated compartment, it may be unsafe to enter because of:

A) High accumulations of nitrogen gas
B) Excess relative humidity
C) A lack of oxygen
D) Built-up electrostatic charges

Correct Answer: C (A lack of oxygen)

Q23. Which of the listed classes of fires can best be extinguished with a water fog?

A) Class A and B
B) Class B and C
C) Class C and D
D) Class A and D

Correct Answer: A (Class A and B)

Q24. The fixed CO2 fire extinguishing system has been activated to extinguish a large engine room bilge fire. When is the best time to vent the combustible products from the engine room?

A) Immediately after the fire goes out
B) After the metal surfaces have cooled down
C) Right before entering the space with an SCBA
D) One hour after initial discharge regardless of temperature

Correct Answer: B (After the metal surfaces have cooled down)

Q25. A definite advantage of using water as a firefighting agent is its characteristic of:

A) Breaking down chemical bonding chains safely
B) Non-conductivity near live machinery
C) Rapid expansion as water absorbs heat and changes to steam
D) Leaving clean residue that acts as a fire barrier

Correct Answer: C (Rapid expansion as water absorbs heat and changes to steam)

Q26. What is NOT a characteristic of a carbon dioxide fire extinguishing agent?

A) It leaves no messy residue after use
B) It is effective even if ventilation is not shut down
C) It displaces air to smother a fire
D) It can cause frostbite upon contact with skin

Correct Answer: B (It is effective even if ventilation is not shut down)

Q27. It is necessary to secure the forced ventilation to a compartment where there is a fire to:

A) Prevent additional oxygen from reaching the fire
B) Retain toxic gases to help smother flames
C) Protect the ventilation fan motor from overspeeding
D) Limit the expansion of cooling steam lines

Correct Answer: A (Prevent additional oxygen from reaching the fire)

Q28. The four basic components of a fire are a chain reaction, heat, fuel, and:

A) Nitrogen
B) Carbon dioxide
C) Oxygen
D) Carbon monoxide

Correct Answer: C (Oxygen)

Q29. The flammable limits of an atmosphere are the:

A) Maximum temperatures at which an oil source produces vapors
B) Upper and lower percentage of vapor concentrations in an atmosphere which will burn if an ignition source is present
C) Safe ambient operating thresholds for storage facilities
D) Minimum moisture concentrations needed to halt spontaneous ignition

Correct Answer: B (Upper and lower percentage of vapor concentrations in an atmosphere which will burn if an igniti

Q30. A fire must be ventilated:

A) To supply additional oxygen to speed up burning
B) To prevent the gases of combustion from surrounding the firefighters
C) Immediately after triggering a fixed gas smothering system
D) To cool down fuel reservoirs below their flashpoints

Correct Answer: B (To prevent the gases of combustion from surrounding the firefighters)

Practice Set 4

Q1. A fire in a radio transmitter would be of what class under NFPA?

A) Class A
B) Class B
C) Class C
D) Class D

Correct Answer: C (Class C)

Q2. Containers are manufactured as per standards set by:

A) IMO
B) SOLAS
C) ISO
D) MARPOL

Correct Answer: C (ISO)

Q3. What, when removed, will assist in the self-extinguishment of a fire?

A) Nitrogen
B) Carbon dioxide
C) Oxygen
D) Argon

Correct Answer: C (Oxygen)

Q4. Classification of fire bulkheads inside accommodation spaces is based on fire resisting capabilities. They are classified as:

A) Class A, B, C
B) Class X, Y, Z
C) Class 1, 2, 3
D) Class H, J, K

Correct Answer: A (Class A, B, C)

Q5. Good housekeeping on a vessel prevents fire by:

A) Improving the speed of bilge pumps
B) Eliminating potential fuel sources
C) Cooling electrical control systems
D) Increasing structural fire boundaries

Correct Answer: B (Eliminating potential fuel sources)

Q6. "Below lower flammable limit" means:

A) The vapor mixture is highly volatile and ready to explode
B) It is a mixture of hydrocarbon gases which is not sufficient to initiate a fire (too lean)
C) The atmosphere is completely saturated with fuel vapors (too rich)
D) There is insufficient structural oxygen available for combustion

Correct Answer: B (It is a mixture of hydrocarbon gases which is not sufficient to initiate a fire (too lean))

Q7. The oxidation process creates:

A) Cooling vapors
B) Heavy scale deposits
C) Heat
D) Free radical inhibitors

Correct Answer: C (Heat)

Q8. Class A divisions are designed to prevent the passage of flame and smoke to the end of a one-hour standard fire test.

A) True
B) False

Correct Answer: A (True)

Q9. An important tool for fire prevention is:

A) Periodic paint applications
B) Cleaning and good housekeeping
C) Keeping internal ventilation doors wide open
D) Increasing high-expansion foam reserves

Correct Answer: B (Cleaning and good housekeeping)

Q10. The tendency or ability of a liquid to vaporize is called:

A) Viscosity
B) Volatility
C) Density
D) Capillarity

Correct Answer: B (Volatility)

Q11. A person may operate an air compressor in which of the following areas on board a tank barge?

A) Inside a cargo tank hatch
B) Pumproom space
C) Generator room
D) Open deck near vapor manifold lines

Correct Answer: C (Generator room)

Q12. The most efficient way of transferring heat in liquids and gases is called:

A) Conduction
B) Convection
C) Radiation
D) Insulation

Correct Answer: B (Convection)

Q13. Fire and lifeboat stations are required to be listed on the:

A) Logbook register
B) Master list (station bill)
C) Cargo manifest layout
D) Crew contract binder

Correct Answer: B (Master list (station bill))

Q14. Radiation is a process of heat transfer where no mass is exchanged and no medium is required.

A) True
B) False

Correct Answer: A (True)

Q15. You can find the location of your abandon ship post by checking the:

A) Deck logbook entries
B) Bridge passage plan
C) Master list
D) Safety equipment certificate booklet

Correct Answer: C (Master list)

Q16. In order to achieve the fire safety objective, which of the following functional requirements are embodied in SOLAS regulations?

A) Division of the ship into main horizontal/vertical zones by thermal and structural boundaries
B) Separation of accommodation spaces from the remainder of the ship by thermal and structural boundaries
C) Protective means of escape and access for firefighting
D) All of the above

Correct Answer: D (All of the above)

Q17. The vessel's Fire Control plan is laid out on which of the following types of plan?

A) Midship structural section
B) General arrangement
C) Lines plan drawing
D) Tank capacity schematic

Correct Answer: B (General arrangement)

Q18. Carbon dioxide is stored under pressure in liquid form inside a fire:

A) Hose connection reel
B) Foam applicator drum
C) Cylinder
D) Storage box enclosure

Correct Answer: C (Cylinder)

Q19. Which statement concerning carbon dioxide is FALSE?

A) It leaves no dynamic residue on machinery components
B) It is safe to use near personnel in a confined space
C) It extinguishes fire by smothering action
D) It can be utilized safely on electrical fires

Correct Answer: B (It is safe to use near personnel in a confined space)

Q20. Which class division is considered capable of preventing the passage of ONLY flame to the end of the first half-hour of a standard fire test?

A) Class A
B) Class B
C) Class C
D) Class D

Correct Answer: B (Class B)

Q21. A fire hose can be used for purposes other than firefighting service when:

A) Washing down cargo decks at sea
B) Used for drills and testing
C) Supplying cooling fluid to a temporary generator
D) Clearing choked scupper valves

Correct Answer: B (Used for drills and testing)

Q22. How many CO2 fire extinguishers ought to be kept in accommodation areas?

A) One per room corridor
B) No extinguishers
C) Two near the galley spaces
D) Four on each deck landing

Correct Answer: B (No extinguishers)

Q23. Portable foam fire extinguishers are designed for use on which classes of fires?

A) Class A and B fires
B) Class B and C fires
C) Class C and D fires
D) Class A and D fires

Correct Answer: A (Class A and B fires)

Q24. The removal of fuel in the firefighting process is known as:

A) Smothering
B) Cooling
C) Starvation
D) Inhibition

Correct Answer: C (Starvation)

Q25. Which of the following classes of fire would probably occur in the engine room bilges?

A) Class A
B) Class B
C) Class C
D) Class D

Correct Answer: B (Class B)

Q26. Why is a CO2 type of fire extinguisher preferred for electrical installations?

A) It provides extensive long-term cooling
B) It expands into a sticky visual foam barrier
C) It leaves no residue and does not damage any equipment
D) It can be safely deployed without isolating power circuits

Correct Answer: C (It leaves no residue and does not damage any equipment)

Q27. Which of the listed extinguishing agents is NOT suitable for fighting a liquid paint fire?

A) Dry chemical powder
B) Carbon dioxide gas
C) Water
D) Mechanical foam spray

Correct Answer: C (Water)

Q28. As per NFPA (National Fire Protection Association), a fire in cooking appliances involving vegetable/animal oils and fats is classified as:

A) Class B
B) Class D
C) Class C
D) Class K

Correct Answer: D (Class K)

Q29. Foam is effective in combating which classes of fire?

A) Class A and B
B) Class B and C
C) Class C and D
D) Class A and D

Correct Answer: A (Class A and B)

Q30. Designated areas for smoking are specified in the:

A) International code manual
B) Company manual
C) Port state entry document
D) Crew mess instruction card

Correct Answer: B (Company manual)

Practice Set 5

Q1. The spreading of fire as a result of heat being carried through a vessel's ventilation system is an example of heat transfer by:

A) Conduction
B) Convection
C) Radiation
D) Absorption

Correct Answer: B (Convection)

Q2. On cargo and miscellaneous vessels, what is NOT a required part of the fireman's outfit?

A) Protective clothing
B) Rigid boots and gloves
C) Combustible gas indicator
D) Flame safety lamp or breathing apparatus

Correct Answer: C (Combustible gas indicator)

Q3. A fire hose kept in its box should have a wrench and an:

A) All-purpose nozzle
B) High expansion foam generator tube
C) International Shore standard coupling flange
D) High-pressure mechanical lance tool

Correct Answer: A (All-purpose nozzle)

Q4. Actuating the fixed CO_2 system should cause the automatic shutdown of the:

A) Main engine propulsion controls
B) Emergency bilge pump circuits
C) Supply and exhaust ventilation
D) Main switchboard power grid distribution

Correct Answer: C (Supply and exhaust ventilation)

Q5. During a fire onboard, communication is to be:

A) Complex, thoroughly detailed, and extensive
B) Concise, precise, and short
C) Relayed solely via messengers
D) Postponed until structural assessments end

Correct Answer: B (Concise, precise, and short)

Q6. A solid stream of water is produced by an all-purpose firefighting nozzle when the handle is:

A) Pushed all the way forward
B) Positioned halfway into its casing
C) Pulled all the way back
D) Twisted ninety degrees counterclockwise

Correct Answer: C (Pulled all the way back)

Q7. Immediately after extinguishing a fire with CO_2, it is advisable to:

A) Restore structural compartment ventilation fully
B) Open up all access ports to inspect inside details
C) Stand by with water or other agents
D) Drain internal oil recovery sumps directly

Correct Answer: C (Stand by with water or other agents)

Q8. Ships carrying more than 36 passengers should maintain a:

A) Continuous automated water system monitor
B) Fire patrol system
C) Double bulkhead smoke containment grid
D) Dedicated fire drill barge trailer connection

Correct Answer: B (Fire patrol system)

Q9. During cargo operations, a deck fire has occurred due to a leaking cargo line. You should first:

A) Rig a mechanical foam generator station
B) Sound the international collision alarm whistle
C) Stop the transfer of cargo
D) Douse structural pipelines with sea spray

Correct Answer: C (Stop the transfer of cargo)

Q10. Radio station logs involving communication during a disaster shall be kept by the station licensee for at least:

A) One year from entry initiation
B) Three years from the date of the last entry
C) Five years under legal maritime seals
D) Ten years inside safe archive storage boxes

Correct Answer: B (Three years from the date of the last entry)

Q11. The deep fryer oil overheated because of a:

A) Lack of fresh liquid fat inputs
B) Thermostat malfunctioning
C) Mechanical failure inside the extractor canopy
D) Blockage inside draining pipe lines

Correct Answer: B (Thermostat malfunctioning)

Q12. The best way to test the Inmarsat-C terminal is to:

A) Place a voice phone link call to shore coordination networks
B) Compose and send a brief message to your own Inmarsat-C terminal
C) Sound the internal automatic distress loop button
D) Contact an adjacent vessel via bridge VHF radio link

Correct Answer: B (Compose and send a brief message to your own Inmarsat-C terminal)

Q13. The control panel of a fire detection system must have all of the following EXCEPT:

A) Visual indicators showing alarm sectors
B) An audible device to signal alarms or system faults
C) A way to bypass the entire panel if it malfunctions
D) Test switches to check structural loop indicators

Correct Answer: C (A way to bypass the entire panel if it malfunctions)

Q14. Tapioca is susceptible to ________ when damp.

A) Chemical crystallization
B) Liquefaction slide effects
C) Spontaneous combustion
D) Electrostatic discharges

Correct Answer: C (Spontaneous combustion)

Q15. As a firefighting medium, CO_2 can be dangerous under certain conditions as it can cause:

A) Severe toxic vapor contamination inside boiler loops
B) Freeze burns and blistering
C) Spontaneous flashover due to compression waves
D) Corrosion of sensitive mechanical gears

Correct Answer: B (Freeze burns and blistering)

Q16. As per NFPA, a fire in an electrical generator is considered to be a:

A) Class A fire
B) Class B fire
C) Class C fire
D) Class D fire

Correct Answer: C (Class C fire)

Q17. If passengers are on board when an abandon ship drill is carried out, they should:

A) Remain inside their cabins behind closed safety doors
B) Gather exclusively at public dining lounges
C) Take part
D) Be evacuated on an auxiliary service tender craft

Correct Answer: C (Take part)

Q18. On the vessel's Fire Control plan, all parts of a fixed fire suppression system are listed EXCEPT:

A) Location of piping distribution loops
B) Total weight and type of chemical storage canisters
C) Instructions for activation of the system
D) Placement of discharge nozzles across compartments

Correct Answer: C (Instructions for activation of the system)

Q19. Through which of the listed processes is sufficient heat produced to cause spontaneous ignition?

A) Heat of condensation
B) Heat of vaporization
C) Heat of oxidation
D) Heat of radiation

Correct Answer: C (Heat of oxidation)

Q20. Fusible link fire dampers are operated by:

A) Electronic loop control commands from the safety desk
B) The heat of the fire melting the link
C) Hydrostatic water pressure changes from water spray systems
D) Structural balance wires linked with the bridge desk

Correct Answer: B (The heat of the fire melting the link)

Q21. You have just extinguished an oil fire on the floor plates of the engine room with a 15-pound CO2 extinguisher. Which of the listed dangers should you now be preparing to handle?

A) Complete systemic failure of main engine gear oil feeds
B) Toxic gas ignition inside air trunk lines
C) Reflashing of the fire
D) Frostbite cracks inside fuel line connection joints

Correct Answer: C (Reflashing of the fire)

Q22. The height of a VHF radio antenna is more important than the power output wattage of the radio because:

A) Skywave atmospheric returns are highly unstable at sea
B) VHF communications are basically line of sight
C) Higher placement reduces tracking ground interference loops
D) Internal wire resistances limit low-level signal paths

Correct Answer: B (VHF communications are basically line of sight)

Q23. Which extinguishing agent is most likely to allow reflash as a result of not cooling the fuel below its ignition temperature?

A) Aqueous film-forming foam (AFFF)
B) CO2
C) Water fog stream
D) Dry chemical powder mix

Correct Answer: B (CO2)

Q24. A vessel's Fire Control plan shall have:

A) Blueprint files locked exclusively in the captain's office
B) A duplicate set of plans permanently stored outside the deck house in a prominent weatherproof enclosure
C) Microfiche summary records mounted inside all lifeboats
D) Plastic layout cards stapled on every accommodation deck door

Correct Answer: B (A duplicate set of plans permanently stored outside the deck house in a prominent weatherproof

Q25. In the regulations for cargo vessels, which statement is true concerning a fireman's outfit?

A) Each outfit must contain an electric welding cutting tool
B) Each fireman's outfit shall contain a flame safety lamp of an approved type
C) Air tanks must carry five hours of continuous storage reserves
D) Two outfits are needed for each active deck crewman

Correct Answer: B (Each fireman's outfit shall contain a flame safety lamp of an approved type)

Q26. Fire alarm system thermostats are actuated by:

A) Air pressure changes inside trunk compartments
B) The difference in thermal expansion of two dissimilar metals
C) Light scattering effects across laser sensors
D) Moisture tracking limits inside structural wiring panels

Correct Answer: B (The difference in thermal expansion of two dissimilar metals)

Q27. For the safety of the personnel working with fire hoses, the fire pumps are fitted with a:

A) Remote automated vacuum seal breaker
B) Mechanical safety clutch linkage on driving engines
C) Pressure gauge and relief valve on the discharge side
D) Flexible expansion joint directly on the water suction intake

Correct Answer: C (Pressure gauge and relief valve on the discharge side)

Q28. Dry chemical extinguishers extinguish Class B fires to the greatest extent by:

A) Cooling fuel temperatures down below flashpoint margins
B) Displacing essential atmospheric nitrogen compounds
C) Breaking the chain reaction
D) Freezing burning vapor pools immediately

Correct Answer: C (Breaking the chain reaction)

Q29. What shall be conducted during a fire and boat drill?

A) Cargo tank hatches must be opened for visual gas checking
B) All watertight doors in the vicinity of the drill shall be operated
C) Emergency bilge suctions must be run on full backflow parameters
D) Lifeboats must be launched and run at service speeds for an hour

Correct Answer: B (All watertight doors in the vicinity of the drill shall be operated)

Q30. When approaching a fire from windward, you should shield firefighters from the fire by using an applicator and:

A) High velocity solid water jet
B) High expansion foam blanket throw
C) Low velocity fog
D) CO2 continuous vapor jet

Correct Answer: C (Low velocity fog)

Practice Set 6

Q1. The extinguishing agent most likely to allow re-ignition of a fire is:

A) Mechanical foam spray
B) Carbon dioxide
C) Water fog
D) Dry chemical powder

Correct Answer: B (Carbon dioxide)

Q2. The risk of fire and explosion is ________ in tankers.

A) Negligible under standard cruising speeds
B) Very high
C) Identical to generic dry cargo vessels
D) Controlled solely by hull thickness layout

Correct Answer: B (Very high)

Q3. Why should foam be banked off a bulkhead when extinguishing an oil fire?

A) To speed up chemical stabilization loops of active foam bubbles
B) To prevent agitation of the oil and spreading the fire
C) To maximize the coverage range of standard delivery equipment
D) To avoid cooling structural steel components unevenly

Correct Answer: B (To prevent agitation of the oil and spreading the fire)

Q4. A large oil fire on the deck of a ship can be fought most effectively with:

A) Solid streams of sea water
B) High pressure carbon dioxide gas grids
C) Foam
D) Portable dry powder spray canisters

Correct Answer: C (Foam)

Q5. The minimum flashpoint of oil fuel carried on board a ship is:

A) 40 degrees Celsius
B) 50 degrees Celsius
C) 60 degrees Celsius
D) 70 degrees Celsius

Correct Answer: C (60 degrees Celsius)

Q6. You are operating a fire hose with an applicator attached. If you put the handle of the nozzle in the vertical position, you will use:

A) High velocity solid jet stream
B) Shutoff position with zero output flow
C) Low velocity fog
D) Foam solution delivery parameter

Correct Answer: C (Low velocity fog)

Q7. To safely enter a compartment where CO2 has been released from a fixed extinguishing system, you should:

A) Wear a generic particulate dust filter mask
B) Run a hand-held exhaust blower for five minutes first
C) Wear a self-contained breathing apparatus (SCBA)
D) Wait ten minutes for gas settlement before walking in

Correct Answer: C (Wear a self-contained breathing apparatus (SCBA))

Q8. The international shore connection:

A) Links ship navigation logs with port administration grids
B) Allows hookup of firefighting water from shore facilities
C) Supplies high voltage electrical power to propulsion links
D) Drains oily bilge slops safely to dockside collection tanks

Correct Answer: B (Allows hookup of firefighting water from shore facilities)

Q9. A fire has broken out on the stern of your vessel. You should maneuver your vessel so the wind:

A) Comes directly over the stern layout beam
B) Blows at right angles to cross-vent the hull decks
C) Comes over the bow
D) Changes into circular thermal swirling patterns

Correct Answer: C (Comes over the bow)

Q10. An extinguishing agent which effectively cools, dilutes combustible vapors, removes oxygen, and provides a heat and smoke screen is:

A) Halon gas
B) Dry chemical powder formulation
C) Water fog
D) Aqueous film-forming foam solution

Correct Answer: C (Water fog)

Q11. The expansion of LFL is:

A) Liquid Fuel Line
B) Low Flashpoint Liquid
C) Lower Flammable Limit
D) Light Flame Location

Correct Answer: C (Lower Flammable Limit)

Q12. Which statement describes the primary process by which fires are extinguished by dry chemical?

A) The dry chemical creates a dense visual foam overlay block
B) The dry chemical powder attacks the fuel and oxygen chain reaction
C) The active agent drops fuel core parameters past absolute freezing margins
D) It relies on heavy moisture transformation into structural vapor screens

Correct Answer: B (The dry chemical powder attacks the fuel and oxygen chain reaction)

Q13. What is the most important consideration when determining to fight an electrical fire?

A) The replacement costs of burnt circuitry components
B) The clean-up time required for active extinguishing residue
C) Danger of shock to personnel
D) Shutting down auxiliary lighting networks across accommodation spaces

Correct Answer: C (Danger of shock to personnel)

Q14. If you see an oil leakage, you clean it immediately. This is an example of:

A) Preventive machinery tuning parameters
B) ISO structural standard declaration rules
C) Good housekeeping
D) Deck log monitoring instructions

Correct Answer: C (Good housekeeping)

Q15. When dry chemical extinguishers are used to put out Class B fires, there is a danger of reflash because dry chemical:

A) Reacts aggressively with marine steel materials
B) Does little or no cooling
C) Displaces air structures for only a few seconds
D) Vaporizes into flammable gas compounds under heat exposure

Correct Answer: B (Does little or no cooling)

Q16. Which of the following fire extinguishing agents has the greatest capacity for heat absorption?

A) Carbon dioxide gas jet
B) Dry chemical powder spray
C) Water fog
D) Mechanical foam blanket expansion

Correct Answer: C (Water fog)

Q17. The expansion of IBC code is:

A) International Bulk Cargo code
B) International Bulk Chemical code
C) Intermaritime Boiler Construction code
D) International Bilge Control code

Correct Answer: B (International Bulk Chemical code)

Q18. As compared to carbon dioxide, dry chemical has which advantage?

A) It leaves no visible residue cleanup footprint on components
B) It offers significant localized thermal cooling effects
C) Greater range
D) It is non-toxic inside confined spaces

Correct Answer: C (Greater range)

Q19. Which of the listed firefighting agents and associated applications provide the largest shielding effect for the firefighter?

A) Straight high-velocity water jet
B) High-expansion chemical foam spray
C) Low velocity water fog
D) Continuous carbon dioxide stream jet

Correct Answer: C (Low velocity water fog)

Q20. An example of a type of smoke detector is the:

A) Thermal differential bimetallic link switch
B) Ionization type
C) Hydrostatic expansion capsule sensor
D) Visual infrared flame scanner screen

Correct Answer: B (Ionization type)

Q21. Which firefighting agent is most effective in removing heat?

A) Carbon dioxide vapor stream
B) Dry chemical powder blanket throw
C) Water spray
D) Mechanical structural foam solution

Correct Answer: C (Water spray)

Q22. When water fog is used as an extinguishing agent, the fire is extinguished principally by the:

A) Displacement of active fuel structures safely
B) Absorption of heat
C) Exclusion of atmospheric nitrogen elements
D) Chemical breakdown of active free radicals

Correct Answer: B (Absorption of heat)

Q23. While fighting a coal fire, forced ventilation can be used to bring down the temperature.

A) True
B) False

Correct Answer: B (False)

Q24. Fire extinguishers of sizes 3, 4, and 5 are designated as:

A) Hand portable
B) Semi-portable
C) Fixed system bank units
D) Auxiliary utility modules

Correct Answer: B (Semi-portable)

Q25. Dangerous goods are classified into how many groups?

A) Three groups
B) Five groups
C) Nine groups
D) Twelve groups

Correct Answer: C (Nine groups)

Q26. Fire safety is achieved by using fire retardant materials.

A) True
B) False

Correct Answer: A (True)

Q27. The discharge from a carbon dioxide fire extinguisher should be directed:

A) At the top perimeter layout of smoke clouds
B) At the base of the flames
C) Directly into adjacent ventilation return trunks
D) In wide looping circles across structural partition bulkheads

Correct Answer: B (At the base of the flames)

Q28. Which statement is true concerning the use of dry chemical extinguishers?

A) You should direct the spray at the top edges of structural smoke screens
B) You should direct the spray at the base of the fire
C) They are deployed safely on metal fires like sodium or magnesium
D) They provide deep cooling to prevent any risk of re-ignition

Correct Answer: B (You should direct the spray at the base of the fire)

Q29. Alcohol-resistant foam is also known as all-purpose foam.

A) True
B) False

Correct Answer: A (True)

Q30. What is the minimum number of portable fire extinguishers that must be carried on ships of 1000 GRT and above?

A) Two
B) Three
C) Five
D) Eight

Correct Answer: C (Five)